

Sustainable Intervention Points: A Framework for Designing High-Leverage Human Roles in AI Pipelines

Eduardo Baena
Northeastern University
Boston, MA, USA
e.baena@northeastern.edu

Gabriela Torres
Aethos
Cambridge, MA, USA
gabriela@aethos.org

Angelica Chincaro
IA Collaborative
Chicago, IL, USA
achincaro@iacollaborative.com

Abstract

When AI is used to automate a workflow, the standard approach to human oversight is to have a human reviewer checking the system’s outputs. This approach has well-documented limitations: reviewers develop automation bias, vigilance declines within minutes, and the skills needed to detect errors degrade over time. The operating environment is also non-stationary: distributions shift, regulations update, and new failure modes emerge. Most research on human-AI collaboration focuses on improving the review interaction itself, through better explanations or calibration aids, without questioning where in the pipeline humans should be placed. We call this placement variable *leverage*: the number of downstream decisions materially changed by a single human action. Our framework defines four tiers of human involvement organized by leverage, from individual output review to strategic constraint specification, and introduces the Sustainable Intervention Point Score (SIPS) to compare configurations across quality, cognitive load, skill preservation, learning signal, and human error amplification. Because the best configuration depends on the specific system and changes as it evolves, the framework is designed for repeated empirical evaluation. We illustrate with credit risk assessment and AI-assisted content management.

CCS Concepts

• **Human-centered computing** → HCI design and evaluation methods.

Keywords

Human-in-the-loop, human-AI alignment, bidirectional alignment, intervention design, leverage, co-evolution, automation bias, sustainable AI

ACM Reference Format:

Eduardo Baena, Gabriela Torres, and Angelica Chincaro. 2026. Sustainable Intervention Points: A Framework for Designing High-Leverage Human Roles in AI Pipelines. In *Proceedings of BiAlign Workshop on Bidirectional Human-AI Alignment (BiAlign@CHI’26)*. ACM, New York, NY, USA, 10 pages. <https://doi.org/10.1145/nnnnnnnn.nnnnnnnn>

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BiAlign@CHI’26, Barcelona, Spain

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ACM ISBN 978-x-xxxx-xxxx-x/YYYY/MM

<https://doi.org/10.1145/nnnnnnnn.nnnnnnnn>

1 Introduction

The deployment of AI systems in high-stakes domains, including healthcare diagnostics [35], financial credit decisions [10], and criminal justice risk assessments [1], has made effective human oversight essential. In each domain, errors carry real consequences, and purely automated systems face both ethical objections and regulatory constraints. The European Union’s AI Act, which entered into force in 2024, mandates human oversight for high-risk AI systems, requiring that such systems “can be effectively overseen by natural persons” [13]. Similar requirements are emerging in other jurisdictions. Yet despite this regulatory push, systematic methods for designing effective oversight remain scarce. The dominant approach, positioning humans as reviewers of AI outputs, may be structurally misaligned with human cognitive capabilities.

1.1 The Problem with Current Approaches

The standard HITL approach routes uncertain cases to human reviewers. This ignores decades of human factors research:

Automation bias. Humans systematically over-rely on automated recommendations, accepting incorrect outputs and missing errors the system missed [16, 34]. This bias is robust across populations, affecting experts and novices equally, and resistant to training interventions.

Vigilance decrement. Sustained monitoring is cognitively demanding, with performance degrading significantly after approximately 15–20 minutes [48]. Systems that require humans to maintain attention while rarely intervening create conditions for failure.

Skill atrophy. When humans are relegated to passive oversight roles, their independent judgment capabilities degrade over time [12]. Recent research demonstrates that AI-induced skill decay may be particularly severe because AI systems are designed to take over cognitive processes, leading to “fewer opportunities to keep skills honed” [30]. This decay can occur without the performer noticing.

These phenomena suggest that the most common HITL configuration, where humans review AI outputs at the end of a pipeline, may be structurally prone to failure. The underlying issue is one of alignment: the system’s design for human involvement is misaligned with the conditions under which humans can perform effectively.

Recent work has begun addressing pieces of this challenge at the interface level. Ma et al. [29] show that replacing passive accept/reject interfaces with deliberative human-AI discussion improves appropriate reliance in decision tasks. Senoner et al. [37] find that explainable AI improves task performance in collaborative settings. Kuper et al. [26] identify psychological factors (risk tolerance, need

for cognition, trust propensity) that moderate reliance on AI clinical decision support. These contributions are valuable, but they operate within the existing paradigm of human-reviews-AI-output. They improve *how* humans interact with AI recommendations at a fixed pipeline position, without addressing *where* in the pipeline the human should be positioned. Our framework addresses this orthogonal design variable.

1.2 The Missing Variable: Leverage

When AI automates a workflow, the critical design question concerns how the human role transforms. Consider two configurations:

- **Configuration A:** AI scores each application; humans review flagged cases ($L \approx 1$).
- **Configuration B:** Humans define risk thresholds, specify prohibited factors, and establish acceptable error rates; AI executes within those boundaries ($L > 1000$).

We call this ratio *leverage*: the downstream reach of a single human intervention. Configuration B repositions the human toward architectural reasoning about risk policy while AI handles routine scoring. The human’s domain expertise is preserved and exercised, directed at system-level quality.

This distinction is largely absent from existing research. Meta-analytic evidence synthesizing over 100 experiments finds significant heterogeneity in outcomes [46], with studies examining pre-determined task delegation showing more promising results than those where humans simply reviewed AI outputs.

1.3 The Persistent Need for Human Involvement

A common assumption is that human involvement is temporary, a stopgap until models improve. This confuses two different problems. A better model resolves more cases, but the operating environment is non-stationary: data distributions shift [2], regulations change [13], user populations evolve, and novel failures emerge from system-level interactions [5]. The operating environment generates these challenges independently of model quality. Production ML systems routinely degrade within months of deployment as input distributions diverge from training data [14], and LLM-based systems add a layer of non-determinism that makes even static environments unpredictable [3].

Model improvement transforms the optimal intervention location while maintaining the need for intervention. An AI content agent with a superior language model still requires someone to define appropriate content as cultural context shifts; a credit scoring model that achieves 99% accuracy on historical data still needs an analyst to recognize when a new lending product creates risk patterns absent from the training set. The dominant failure mode is environmental change that the model cannot detect on its own. Our framework treats the intervention point as a variable that must be re-optimized as the system and its environment co-evolve.

1.4 Contributions

We make four contributions:

- (1) **Theoretical framework:** We formalize HITL design as a dual-variable optimization problem, jointly considering technical pipeline characteristics and human behavioral dynamics.
- (2) **Taxonomy of intervention types:** We propose four tiers of intervention organized by leverage, from strategic control (highest) to mechanical review (lowest).
- (3) **Quantifiable metrics:** We introduce the Sustainable Intervention Point Score (SIPS), a composite metric grounded in established measurement frameworks including NASA-TLX for cognitive load [19], RSR/RAIR for appropriate reliance [36], and skill retention protocols from human factors research [23].
- (4) **Experimental methodology:** We propose a five-phase protocol for identifying optimal intervention points empirically, including longitudinal validation and a simulation-based path toward cross-domain generalization.

2 Background and Related Work

Our framework synthesizes research across seven domains. We highlight the findings most relevant to intervention point design.

AI system dynamics. Production AI systems exhibit concept drift [2, 22], model degradation [14], non-determinism (5–15% accuracy variation at temperature=0 [3, 20]), and emergent failures [5]. These properties make human involvement a structural requirement. Kreuzberger et al. [25] establish that ML models require continuous validation through MLOps practices.

Human labor in AI systems. Ethnographic research reveals invisible human labor throughout AI systems [17, 44]. Chandhiramowuli et al. [9] document pervasive but invisible counting and quantification regimes governing annotation labor, challenging narratives of AI as autonomous.

Human-AI complementarity. Hemmer et al. [21, 43] show that complementarity is rarely achieved in practice. Steyvers et al. [41] demonstrate it requires sufficiently uncorrelated errors. Meta-analytic evidence [46] confirms significant heterogeneity, with predetermined task delegation showing more promise than passive human review. These findings support our focus on *where* humans intervene.

Trust and reliance dynamics. Trust in automation evolves through interaction [27] and mediates reliance behavior [7]. Mehrotra et al. [32] review 96 empirical studies, finding systematic miscalibration: low-reliability systems are over-trusted while high-reliability systems are under-trusted, with direct implications for HITL design. Trust dynamics can be modeled mathematically via Decision Field Theory [15]. Schemmer et al. [36] introduced metrics for appropriate reliance (RSR and RAIR) that capture whether humans correctly override AI errors and correctly follow AI when it is right, going beyond simple override rates. One finding stands out: explanations increase perceived acceptability but “are often insufficient to improve decision accuracy or mitigate automation bias” [36], suggesting structural changes to *how* humans are involved may be more productive than interface-level interventions alone. Sterz et al. [40] identify four necessary conditions for effective oversight (means for intervention, access to information,

agency to override, and aligned intentions) that are often violated in deployed systems.

Cognitive load and skill preservation. The NASA Task Load Index (NASA-TLX) [19] is the dominant cognitive workload measure in human factors research. High cognitive load correlates with vigilance decrement [48] and increased error rates, making cognitive load a hard constraint on intervention design. Skill decay research [23] meta-analyzing 53 studies shows cognitive skills degrade faster than physical skills. Macnamara et al. [30] extend this concern, arguing that AI-driven deskilling is acute when the automated tasks are cognitive. Casner et al. [8] demonstrated this empirically with pilots: procedural skills remained largely intact after extended autopilot use, but cognitive skills (situation awareness, planning, failure recognition) showed significant declines. Intervention designs must therefore be evaluated for cognitive skill maintenance in addition to immediate performance.

Control systems models. McRuer’s crossover model [31] and the Optimal Control Model [24] provide quantifiable human operator transfer functions. Recent work formalizes LLM systems as discrete stochastic dynamical systems [6, 28], enabling a unified control-theoretic treatment of both AI pipelines and human operators.

Agentic AI oversight. For agentic systems [47], per-action approval eliminates efficiency gains. Overman et al. [33] model human-AI oversight as a two-player Markov Game. Human oversight must shift from reviewing individual actions to specifying goals, setting constraints, and evaluating outcomes, a structural shift toward higher-leverage intervention points.

3 A Framework for Intervention Point Design

3.1 Leverage

We define the *leverage* of an intervention point as the downstream reach per intervention: the number of decisions materially changed by human input at that point. Leverage measures propagation, whereas the quality term $Q(p)$ (introduced below) captures whether that propagation improves or degrades system performance.

Formally, leverage is estimated empirically over an observation window W :

$$\hat{L}_W(p) = \frac{N_W^{\text{chg}}(p)}{N_W^{\text{int}}(p)} \quad (1)$$

where $N_W^{\text{int}}(p)$ is the number of human interventions at point p during W , and $N_W^{\text{chg}}(p)$ is the number of downstream decisions materially changed by those interventions. A decision is materially changed if the intervention alters the decision outcome, the feasible action set, or the decision rule applied later in the pipeline. For brevity, we write $L(p)$ when the observation window is implicit.

High-leverage interventions (defining constraints, curating training data, identifying systematic errors) propagate to many downstream decisions per human action. Low-leverage interventions (reviewing individual outputs) propagate to one decision each.

3.2 Taxonomy of Intervention Types

We propose four tiers organized by leverage, with illustrative ranges representing approximate orders of magnitude:

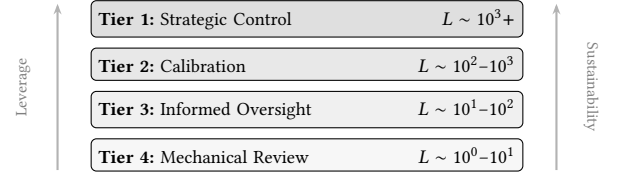


Figure 1: Four-tier taxonomy of human intervention types. Higher tiers yield greater downstream reach per human action and better sustainability over time.

Tier 1: Strategic Control ($L \sim 10^3$ or higher). Humans define objectives, establish hard constraints, and specify operating boundaries. Each decision shapes the entire system’s behavior.

Tier 2: Calibration ($L \sim 10^2-10^3$). Humans label edge cases, identify drift, diagnose systematic errors. Each intervention improves many future outputs.

Tier 3: Informed Oversight ($L \sim 10^1-10^2$). Humans review samples or exceptions with potential for pattern identification. Value derives from identifying patterns across outputs.

Tier 4: Mechanical Review ($L \sim 10^0-10^1$). Humans review individual outputs without pattern identification. Most susceptible to automation bias and vigilance decrement.

3.3 The Sustainable Intervention Point Score (SIPS)

Beyond immediate performance, intervention configurations should be evaluated for sustainability, the ability to maintain effectiveness over time. We propose a composite metric grounded in established measurement frameworks, integrating six components:

Quality (Q): Output quality improvement relative to baseline, measured through domain-specific accuracy metrics following standard practice in human-AI collaboration research [46].

Leverage (L): Downstream reach per intervention as defined above. While novel to HITL research, this concept parallels “span of control” in organizational theory and “influence” in network analysis.

Learning Signal (λ): The degree to which human interventions generate data that enables system improvement. Operationalized as the proportion of interventions that identify previously unknown failure modes, provide corrective labels usable for retraining, or reveal distribution drift, drawing on active learning literature [38].

Cognitive Load (γ): Measured via NASA-TLX [18, 19] on its standard 0–100 scale. We normalize cognitive load in the SIPS formula as $1 + \gamma/100$ to map the range to $[1, 2]$, avoiding a singularity at zero load while ensuring that maximum cognitive load halves the score.

Skill Preservation (σ): The trajectory of human capabilities over time, measured through periodic unassisted performance testing following protocols from skill decay research [23]. We allow bounded degradation: if skill level remains above a retraining threshold $\sigma_{\min}$, the configuration is viable.

Human Error Penalty (ϵ_h): The rate at which human interventions introduce errors. High-leverage intervention points amplify both correct and incorrect human decisions; this penalty ensures

SIPS penalizes configurations where erroneous human input propagates through the system at scale.

3.3.1 SIPS Formulation. A sustainable intervention point satisfies five criteria:

- (1) **Quality:** Improves output quality relative to baseline ($Q > 1$).
- (2) **Learning:** Captures signal that enables system improvement ($\lambda > 0$).
- (3) **Preservation:** Maintains human capabilities above a retraining threshold ($\sigma \geq \sigma_{\min}$).
- (4) **Feasibility:** Remains within human cognitive capacity ($\gamma < \gamma_{\max}$).
- (5) **Reliability:** Human error rate remains bounded ($\epsilon_h < \epsilon_{\max}$).

The composite metric:

$$\text{SIPS}(p) = \frac{Q(p) \times L(p) \times \lambda(p) \times (1 - \epsilon_h(p))}{1 + \gamma(p)/100} \times S(\sigma(p)) \quad (2)$$

where $S(\sigma)$ is a soft preservation function that smoothly penalizes skill degradation while allowing bounded decline above a retraining threshold $\sigma_{\min}$:

$$S(\sigma) = \begin{cases} 1 & \text{if } \sigma \geq \sigma_{\text{base}} \\ \frac{\sigma - \sigma_{\min}}{\sigma_{\text{base}} - \sigma_{\min}} & \text{if } \sigma_{\min} \leq \sigma < \sigma_{\text{base}} \\ 0 & \text{if } \sigma < \sigma_{\min} \end{cases} \quad (3)$$

We chose a multiplicative form so that any component at zero collapses the score: a configuration with no learning signal ($\lambda = 0$), total human error ($\epsilon_h = 1$), or zero quality improvement ($Q = 0$) scores zero regardless of leverage. An additive form would allow high leverage to compensate for these failures, which we consider unacceptable. The denominator normalizes cognitive cost: maximum TLX load ($\gamma = 100$) halves the score relative to a zero-load task. The soft preservation function $S(\sigma)$ applies a smooth penalty for skill degradation while still excluding configurations that push skills below the retraining threshold. When skills degrade to $\sigma_{\min}$, a retraining intervention is triggered to restore capability, making the constraint recoverable.

3.3.2 Appropriate Reliance Integration. SIPS can be augmented with appropriate reliance metrics when comparing configurations at similar leverage levels:

$$\text{SIPS}_{AR}(p) = \text{SIPS}(p) \times \frac{\text{RSR}(p) + \text{RAIR}(p)}{2} \quad (4)$$

This variant penalizes over-reliance (low RSR) or under-utilization of correct AI advice (low RAIR).

3.4 Structured Empirical Comparison

SIPS operates through empirical comparison. Since the human behavioral response $H(p, i)$ must be observed empirically, the framework computes SIPS across tested configurations and selects the best feasible option. We formalize this comparison over two variable classes:

Technical Variables: Pipeline stage p , leverage $L(p)$, rollback capability $R(p)$, AI failure rate $F(p)$, drift rate $D(p, t)$.

Human Variables: Automation bias $A(i, p)$, appropriate reliance $\rho(i)$, cognitive load $\gamma(i)$, skill trajectory $\sigma(i, t)$, learning signal $\lambda(i)$, human error rate $\epsilon_h(i, p)$.

Table 1: Mapping SIPS components to model operations infrastructure

SIPS Component	Operations Infrastructure
$Q(p), F(p)$	Evaluation pipelines, RAG precision/recall, hallucination detection
$D(p, t)$	Distribution monitoring, embedding drift, degradation alerts
$\lambda(p)$	RLHF pipelines, fine-tuning curation, relevance labeling
$\epsilon_h(p)$	Ground-truth test sets, A/B testing, canary deployments

$$\max_{p \in P, i \in I} \frac{Q(p, i, H(p, i)) \cdot L(p) \cdot \lambda(p, i) \cdot (1 - \epsilon_h(p, i))}{1 + \gamma(p, i)/100} \cdot S(\sigma(p, i)) \quad (5)$$

subject to constraints on cognitive load ($\gamma < \gamma_{\max}$), skill preservation ($\sigma \geq \sigma_{\min}$), appropriate reliance ($\rho > \rho_{\min}$), and human error ($\epsilon_h < \epsilon_{\max}$). The constraints serve as hard safety bounds that exclude configurations before the score is compared. The result is a ranking of tested configurations, analogous to how A/B testing selects between design alternatives. SIPS is a design lens that surfaces the tradeoffs between variables (quality vs. cognitive load, leverage vs. error amplification); the formalization makes the interactions between these forces explicit and comparable across configurations.

3.5 Grounding in Model Operations Infrastructure

SIPS requires continuous measurement of its technical variables in production. This infrastructure is precisely what MLOps [25, 42] and AgentOps [11] provide: evaluation pipelines feed $Q(p)$ and $F(p)$; distribution monitoring feeds $D(p, t)$ and triggers SIPS re-evaluation; RLHF and fine-tuning pipelines are the learning signal $\lambda(p)$; and ground-truth test sets enable measurement of $\epsilon_h(p)$.

This relationship is bidirectional: model operations infrastructure makes SIPS measurable, while SIPS tells model operators *where* human feedback and labeling effort generate the highest return. This addresses a gap in current MLOps frameworks, which specify what to monitor without specifying how to allocate human effort across monitoring tasks.

4 Research Questions and Hypotheses

Research Questions

- RQ1:** How does the *leverage* of an intervention point moderate human-AI team performance?
- RQ2:** What configurations enable *sustainable* human involvement, maintaining quality, reliance, and skill preservation over time?
- RQ3:** To what extent are optimal configurations *context-dependent*, varying across domains, tasks, and populations?

Together, these questions reframe the design problem: they ask *where* involvement creates the greatest value per unit of cognitive

effort, and whether that value holds up as both the AI system and the human operator evolve. We propose five testable hypotheses:

H1: Leverage moderates human-AI performance. High-leverage interventions (Tiers 1–2) will produce better outcomes than low-leverage interventions (Tier 4), controlling for total human effort. We predict a non-linear relationship where quality per unit of effort increases sharply at higher leverage tiers.

H2: Active input reduces automation bias. Intervention designs requiring active human reasoning will show higher appropriate reliance rates as measured by RSR and RAIR.

H3: Optimal configurations are context-dependent. The specific intervention point that maximizes SIPS will vary across domains, populations, and interface designs, so universal recommendations are unlikely.

H4: Sustainable intervention points exist. For most AI pipelines, there exist configurations that simultaneously improve quality, capture learning signal, and preserve human capabilities over time.

H5: Tier 4 interventions exhibit quantifiable degradation dynamics. Prior work has established that passive monitoring degrades performance [8, 30]; the rate and drivers of this degradation in production HITL systems remain unmeasured. We hypothesize that degradation follows a predictable curve whose parameters (onset latency, decay rate, asymptote) can be estimated from early observations, enabling proactive reconfiguration.

5 Illustrative Cases

Our framework applies to any AI-automated workflow where human involvement must be deliberately designed. We illustrate with two contrasting domains.

5.1 AI-Assisted Content Management

Consider an AI agent that automates social media content creation. The leverage spectrum applies directly:

At Tier 4, a human approves or rejects each generated post. This configuration either negates the agent’s throughput advantage or, under batch approval, risks publishing harmful content (e.g., brand-damaging statements, factual errors, tone-deaf messaging) that the human failed to detect due to vigilance decrement. The human’s cognitive resources are consumed by repetitive evaluation of individual outputs, with no mechanism for systematic quality improvement.

At Tier 1, the human defines brand voice guidelines, specifies prohibited topics, curates the data sources the agent ingests, and monitors aggregate engagement and sentiment metrics, controlling the *system that produces content*. The agent handles routine generation and scheduling; the human’s domain expertise is directed toward strategic decisions about information management, prompt architecture, and drift monitoring as audience dynamics evolve.

This shift captures our central claim: multiplying output volume while preserving meaningful human control over quality and risk. The human at Tier 1 exercises higher-order judgment (brand strategy, risk assessment, audience analysis) and can rapidly recalibrate when audience behavior shifts or new competitors emerge.

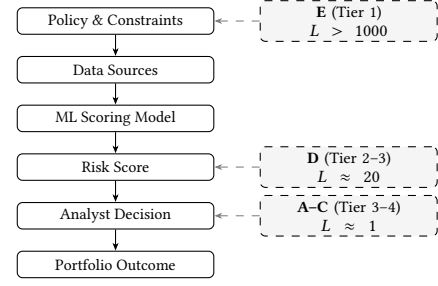


Figure 2: Credit risk pipeline with experimental conditions A–E mapped to intervention points at different leverage tiers.

5.2 Credit Risk Assessment

Current deployments present analysts with an AI-generated risk score plus explanatory features, asking them to approve or override. This configuration suffers from known problems: anchoring from the pre-computed score [45], premature closure of evidence search, declining override rates even when the model makes systematic errors, and progressive loss of independent assessment skills. From a control systems perspective, this is a poorly designed feedback loop: the human receives a pre-computed conclusion, reducing the analyst’s role to passive confirmation.

We propose five experimental conditions spanning all four tiers of the intervention taxonomy, varying both leverage and information presentation:

Condition A (Baseline): AI provides risk score + explanations; human approves/overrides. *Tier 4*, $L \approx 1$.

Condition B (Evidence-on-Demand): AI provides no score. Human requests evidence categories and renders independent judgment; AI score revealed post-decision for calibration. *Tier 3*, *active reasoning*.

Condition C (Blinded Score): AI score masked until human registers preliminary decision, then revealed for calibration. *Tier 3*, *debiasing*.

Condition D (Hybrid Escalation): AI auto-resolves high-confidence cases (>95% confidence). Human reviews uncertain cases plus 5% random sample for drift detection. *Tier 2–3*, $L \approx 20$.

Condition E (Strategic Control): Human defines risk thresholds per product category, specifies prohibited factors, establishes acceptable false-positive/negative rates. AI executes all individual decisions within these boundaries. *Tier 1*, $L > 1000$.

5.2.1 Mapping to Hypotheses and Predicted Outcomes. Conditions A through E span the full leverage spectrum, enabling direct tests of each hypothesis. For H1, we predict Condition E achieves comparable or better portfolio-level accuracy with an order of magnitude less analyst time, because each strategic decision shapes thousands of downstream outcomes. For H2, Conditions B and C should show higher RSR than Condition A, because analysts engage in independent reasoning before seeing the AI conclusion. For H3, we predict senior analysts perform optimally in Condition E (strategic control leverages experience), while juniors benefit from the scaffolding in Condition C. For H4, longitudinal comparison of A vs. B tests whether evidence-on-demand preserves skills better than

Table 2: Operationalization of SIPS components for credit risk assessment

Component	Metric	Measurement
Quality (Q)	Decision accuracy	Loss rate on approved loans False decline rate
Leverage (L)	Decisions materially changed	Cases per analyst-hour
Learning (λ)	System improvement	Drift detection rate Edge cases for retraining
Cog. Load (γ)	NASA-TLX	Post-session questionnaire Time per decision
Skill (σ)	Unassisted accuracy	Bi-weekly held-out test
Error (ϵ_h)	Human errors	Ground-truth comparison
Reliance	RSR	Override rate on AI errors
	RAIR	Accept rate on AI correct

score-first presentation. For H5, Condition A should show declining detection rates with a 2–4 week onset latency.

Credit risk fits this evaluation well because: (1) ground truth is eventually observable (loan performance), (2) regulatory requirements mandate human oversight, (3) the cost structure is well-defined (false approvals vs. false declines), and (4) existing deployments provide baseline data.

6 Proposed Methodology

We propose a five-phase experimental methodology for identifying optimal intervention points in a specific deployment context. The methodology is designed to be domain-agnostic in structure while producing domain-specific results.

Phase 1: Pipeline Mapping. For a target AI system, identify all feasible intervention points, that is, locations in the pipeline where human input could modify system behavior. Characterize each by leverage estimate, current AI failure rate, rollback capability, and drift indicators. This phase produces a candidate set of intervention points for empirical evaluation. For complex systems with many potential points, leverage estimates can be used to prune the set to a tractable number of candidates.

Phase 2: Baseline Measurement. For each candidate intervention point, measure: override rate, appropriate reliance (RSR and RAIR), cognitive load (NASA-TLX), human error rate (ϵ_h), and time per intervention. This phase establishes the empirical parameters needed to compute SIPS. Measurements should be collected under realistic operating conditions to ensure ecological validity for predicting real-world sustainability.

Phase 3: Known-Error Testing. Following established methods for measuring automation bias [4], construct test sets with known AI errors and insert them into normal workflow (10–20% of cases). Measure detection rate per configuration. This phase directly measures the human’s ability to detect AI failures at each intervention point, the core mechanism through which human oversight adds value. Interpretation: detection rate below 50% indicates automation bias dominates; above 80% indicates appropriate reliance.

Phase 4: Configuration Optimization. Compute SIPS for each configuration. Filter by constraints (cognitive load thresholds, minimum detection rates, maximum human error rates). Identify Pareto-optimal configurations balancing quality, sustainability, and cost. Where multiple configurations meet all constraints, the leverage-weighted SIPS ranking provides a systematic basis for selection.

Phase 5: Longitudinal Validation. Monitor selected configurations over 8+ weeks to assess sustainability. Weekly: output quality, detection rate, cognitive load, human error rate. Bi-weekly: skill preservation through unassisted testing. Monthly: system learning through AI performance on held-out sets. Success criterion: quality and detection rate variance below 15%; skills above retraining threshold $\sigma_{\min}$. This phase is what distinguishes our methodology from one-shot evaluations: many intervention configurations that appear effective in short-term testing degrade over weeks as automation bias accumulates and skills atrophy.

6.1 Data Collection and Benchmark Design

Implementing this methodology requires structured data collection at each phase. We outline the practical requirements.

Interaction logs. Each intervention must be recorded with timestamps, available information, the decision made, and the AI’s recommendation (if visible). Existing MLOps platforms (MLflow, Weights & Biases, LangSmith) capture model inputs/outputs; the additional requirement is instrumenting the human decision interface to the same level of detail.

Known-error test sets. Phase 3 requires curated datasets where the AI produces incorrect outputs with known ground truth. These must be domain-specific and realistic enough that participants cannot distinguish them from normal workflow. We recommend a minimum of 50 known-error cases per condition, refreshed quarterly to prevent memorization.

Longitudinal measurement protocol. Phase 5 requires repeated measures over 8+ weeks. We propose bi-weekly “unassisted sessions” (20–30 cases without AI) to track skill trajectories, plus weekly NASA-TLX questionnaires at consistent times to control for circadian and workload confounds.

Ground truth and proxies. Each domain requires a ground truth strategy. In credit risk, loan performance provides delayed ground truth; in content management, engagement metrics serve as proxies. Phase 1 should document the ground truth timeline. To enable cross-study comparison, future work should develop standardized HITL benchmarks with known AI error distributions and calibrated simulation parameters.

Ethical considerations. Longitudinal studies require IRB protocols addressing the tension between performance measurement and ecological validity. Results should be aggregated at the configuration level to avoid punitive use of individual data.

7 Discussion

7.1 Leverage Points as Dynamic Variables

A likely objection is that optimal intervention points depend heavily on model capabilities, pipeline architecture, and data environment. We see this context-dependence as the core problem the framework is designed to address.

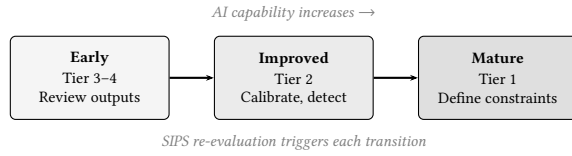


Figure 3: Dynamic leverage shift: as AI capabilities improve, the optimal human intervention moves from output review toward strategic control.

Consider a RAG system for customer support. If retrieval returns irrelevant documents for 30% of queries, the highest-leverage intervention is at Tier 2: curating the knowledge base to improve retrieval across all responses. After fine-tuning reduces hallucination rates, the failure modes shift to compliance violations, and the optimal intervention moves to Tier 1: specifying output constraints and compliance boundaries.

Throughout this progression, human involvement remains necessary but shifts in character. Each model improvement shifts the optimal intervention point (Figure 3). This illustrates three properties our framework handles:

- (1) **Leverage points shift as AI capabilities evolve.** A model improvement that resolves one failure mode exposes others. Static HITL designs cannot adapt; Phase 5 longitudinal monitoring detects shifts through changes in SIPS component values.
- (2) **The framework is a diagnostic methodology.** We provide measurement instruments that identify the optimal configuration for a *specific* deployment at a *specific* lifecycle point, analogous to control systems engineering, which provides theory to design a controller for each plant.
- (3) **Re-evaluation is built in.** When the AI system is updated, Phases 1–4 are re-executed at lower cost because measurement infrastructure is already in place.

7.2 Role of Simulation

A practical limitation is that identifying optimal intervention points requires domain-specific empirical evaluation, a costly 1:1 ratio between experiments and application domains. Simulation cannot replace this empirical requirement: as noted above, the human behavioral response $H(p, i)$ must be observed, and H3 predicts that optimal configurations are context-dependent. Simulating human decision-making from transferred parameters would undermine the framework’s empirical foundation.

Simulation is useful, however, for a narrower set of questions that complement empirical evaluation:

Sensitivity analysis. Once SIPS components have been measured empirically, simulation can assess how robust a configuration ranking is to measurement error. For example, if leverage at a Tier 2 intervention is estimated as $L \approx 200$ with uncertainty ± 80 , simulation can determine whether the ranking of configurations changes across the plausible range. This is particularly valuable because L spans orders of magnitude and dominates SIPS; small estimation errors in leverage can change which configuration appears optimal.

What-if scenarios. Simulation can explore how SIPS would change under hypothetical shifts, such as a model upgrade that halves the AI failure rate or a regulatory change that introduces new compliance requirements. These scenarios help practitioners anticipate when re-evaluation will be needed, without requiring a new empirical study for each hypothetical.

Error amplification analysis. High-leverage configurations amplify human errors (ϵ_h) across many downstream decisions. Simulation can model how different ϵ_h rates propagate through the pipeline at each tier, making the risk of high-leverage error amplification concrete before deployment.

These uses treat simulation as an analytical tool applied to empirically grounded parameters, preserving the framework’s commitment to observation over derivation.

7.3 Implications for Practice

Our framework suggests several practical recommendations:

Shift human effort upstream. Involve humans in defining constraints, curating training data, and identifying systematic errors.

Design for active engagement. Require active reasoning (e.g., withholding AI recommendations until the human forms an independent assessment) to reduce automation bias.

Measure appropriate reliance beyond override rate. Implement known-error testing to assess whether humans detect AI errors.

Monitor longitudinally. Budget for ongoing SIPS measurement as an operational cost. Short-term experiments miss degradation effects that emerge over weeks. At Tiers 1–2, monitor human error amplification (ϵ_h) carefully, since high-leverage interventions amplify both correct and incorrect decisions.

Accept context-dependence. Invest in empirical identification for each deployment context.

7.4 Connection to Bidirectional Human-AI Alignment

Our framework connects directly to recent work on bidirectional alignment [39]. Traditional alignment is unidirectional: configuring AI to match human values at deployment time. Our framework operationalizes *bidirectional co-adaptation* through three mechanisms:

AI→Human adaptation. The AI system’s capabilities and failure modes determine which human tasks create the most value. As models improve, the optimal role shifts from case-level review toward strategic control; SIPS detects when this shift is warranted.

Human→AI adaptation. Tier 1 interventions (constraint specification) directly modify the AI’s behavioral envelope; Tier 2 interventions (calibration, drift diagnosis) generate labeled data that drives model improvement. The learning signal λ quantifies this feedback channel.

Sustainability as alignment maintenance. SIPS tracks whether the human-AI relationship remains viable over time. When SIPS degrades, it signals that alignment has drifted, triggering re-evaluation. An alignment process that degrades human capacity to steer is self-defeating regardless of short-term performance.

7.5 Limitations

Our framework has several limitations that future work should address:

Empirical validation needed. The quantitative predictions (e.g., non-linear leverage-quality relationships, 2–4 week degradation onset) are derived from prior literature and have not yet been tested directly with SIPS. The illustrative cases demonstrate face validity, but testing across diverse domains is required.

Leverage estimation challenges. Quantifying leverage precisely may be difficult in complex pipelines with multiple feedback loops. Because L spans several orders of magnitude while other SIPS components are bounded in $[0, 1]$, leverage dominates cross-tier comparisons; SIPS is most informative when comparing configurations at similar leverage levels.

Sample size at high-leverage tiers. Tier 1 interventions are infrequent by design (e.g., 10 constraint-setting decisions per quarter). Estimating ϵ_h and λ reliably from so few observations is difficult. Similarly, λ is mechanically correlated with L : a single useful intervention out of 10 gives $\lambda = 0.1$, while the same absolute count in 1000 Tier 4 reviews gives $\lambda = 0.001$. Future work should explore confidence-weighted SIPS variants that account for estimation uncertainty.

Individual differences. Optimal configurations likely vary across individuals based on expertise, cognitive style, and trust propensity. A Tier 1 configuration effective for an experienced analyst may produce high ϵ_h for a junior one.

Organizational factors. Our framework focuses on individual human-AI interaction and does not address organizational dynamics, incentive structures, or team configurations that constrain which interventions are feasible in practice.

Domain specificity. Parameters must be determined empirically for each application domain; the framework provides structure but not universal values.

Adversarial scenarios. The human error penalty (ϵ_h) captures unintentional errors, but deliberate manipulation at high-leverage points poses a distinct threat requiring additional safeguards (e.g., multi-person oversight, audit trails) beyond what SIPS currently specifies.

7.6 Open Questions for Workshop Discussion

Our framework raises several questions that suit interdisciplinary workshop discussion:

How should leverage be weighted against equity? High-leverage intervention points concentrate decision-making power in fewer human hands. A Tier 1 configuration where one person’s constraint specification governs thousands of outcomes is efficient but raises questions about accountability, representativeness, and the distribution of decision authority. How should frameworks for bidirectional alignment account for the concentration of human influence at high-leverage points?

When does co-adaptation become co-dependence? Our framework assumes that ongoing human involvement is desirable, but some system trajectories may lead to situations where neither the human nor the AI can function effectively without the other, a form of mutual lock-in that constrains future design choices. How

should alignment frameworks distinguish healthy co-adaptation from problematic co-dependence?

Can SIPS be extended to multi-stakeholder settings? Our current formulation considers a single human operator interacting with an AI system. In practice, AI systems serve diverse stakeholders with potentially conflicting values. Extending SIPS to account for multiple simultaneous alignment relationships (between the AI and different user groups, regulators, and affected populations) remains an open challenge.

8 Conclusion

AI automation multiplies throughput while transforming what human judgment should be directed toward. Our framework provides a structured methodology for designing this transformation: identifying configurations where human expertise creates the greatest downstream impact while remaining cognitively sustainable.

The leverage-based taxonomy and SIPS metric offer concrete tools for comparing intervention configurations, grounded in validated measurement frameworks. The inclusion of human error amplification and soft skill preservation addresses practical risks that prior HITL frameworks have not formalized. Connecting the framework to model operations infrastructure (MLOps/AgentOps) grounds it in observability systems that production deployments already use.

The core contribution is a shift in framing: from designing static oversight to engineering *how* human roles co-evolve with AI capabilities. Productivity gains should not come at the cost of quality, adaptability, or the human capabilities needed to manage systems whose operating environments keep changing. We see this as what bidirectional human-AI alignment looks like in practice: a continuous process of mutual adaptation guided by structured measurement.

Two aspects distinguish this work from prior HITL research. First, the framework treats the human role itself as a design variable. Most existing approaches ask how to help humans do their current task better: clearer explanations, better calibration, reduced bias. Ours asks whether they are doing the right task. Second, SIPS makes the tradeoff between short-term performance and long-term sustainability explicit and quantifiable.

Several extensions are worth pursuing. Production deployments often involve teams at different tiers simultaneously, a senior analyst setting Tier 1 constraints while juniors handle Tier 3 exceptions. Modeling these multi-tier configurations requires extending the optimization to account for coordination costs between roles. A second direction is automated transition detection: the monitoring infrastructure could trigger role changes when SIPS components cross predefined thresholds, replacing periodic re-evaluation with continuous adaptation. Finally, SIPS could serve a regulatory function. The EU AI Act mandates human oversight of high-risk systems but provides no measurement standard; a validated SIPS profile could offer auditable evidence that oversight is both effective and sustainable.

Acknowledgments

We thank Kimberly Chung for her detailed review and comments, which helped improve this paper.

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